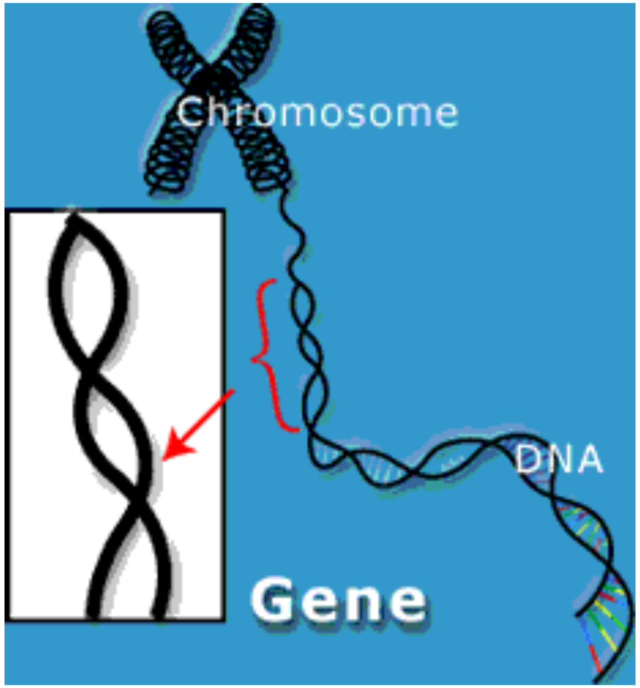
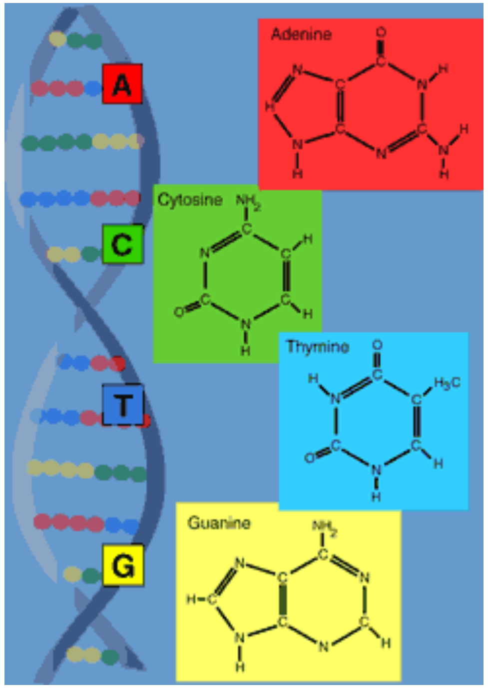
Visual Analysis of Multivariate Biological Networks

Information Visualization (186.141)
TU Vienna, Austria
May 13, 2026

2.1 Introduction

- <u>Correlations:</u> Genome – Chromosome – Gene – DNA

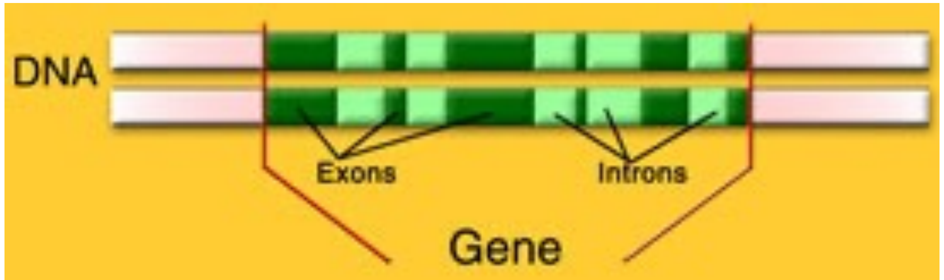




- <u>DNA:</u> Double helix with nucleotides from sugar, phosphate and four bases (Adenin, Cytosin, Guanin, Thymin)

2.1 Introduction

- Genome: only 5% genes consisting of Exons and Introns

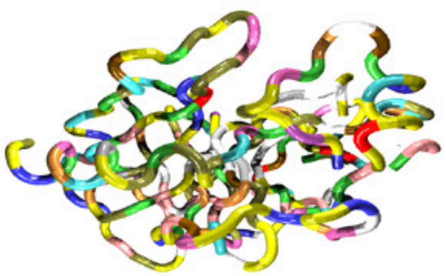
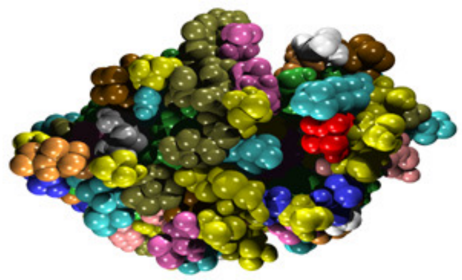


Organism	Base Pairs	Genes
yeast	12 mill.	6000
fruit fly	137 mill.	13.601
human	3 billions	~30.000

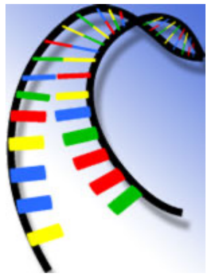
- RNA: Uracil instead of Thymin, single strands

2.1 Introduction

- Proteins are built by amino acids [Video]
- 20 amino acids: coded by nucleotide triplets (Codons)
  - e.g. GCA, GCC, GCG → Alanin

Prime-	Secondary-	Tertiary Structure
ADMVIKAPAG AKVTKAPVAF SHKGHASMDC		

- Protein synthesis
  - Transcription: DNA → mRNA
  - Translation: mRNA → amino acids → protein



http://de.wikipedia.org/wiki/Codon

2.1.1 Tasks of Bioinformatics

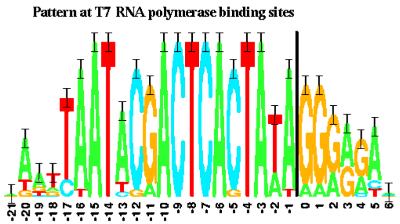
- Support of biologists within processing, analysis and interpretation of large data sets
- Approaches:
  - Biological data bases
  - Methods for comparison and function prediction of sequences
  - Function prediction of proteins
  - Locating of new correlations
  - Simulation of biological processes

2.1.2 Visualizations

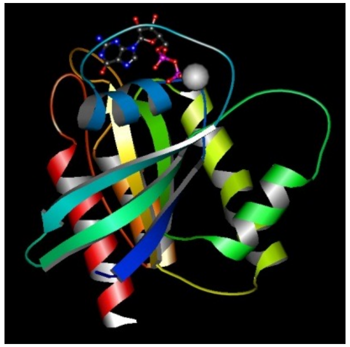
- Visualizations of networks and related structures
  - Biological networks
  - Phylogenetic trees
  - RNA-secondary structure

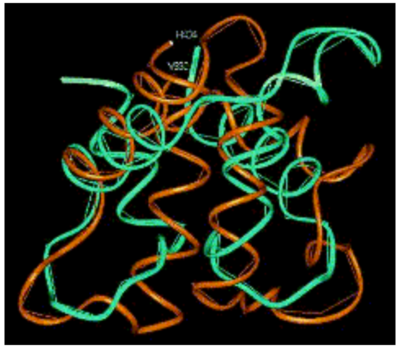
Here, we will focus on this issue only!

- Visualizations of sequences
  - Sequence alignments

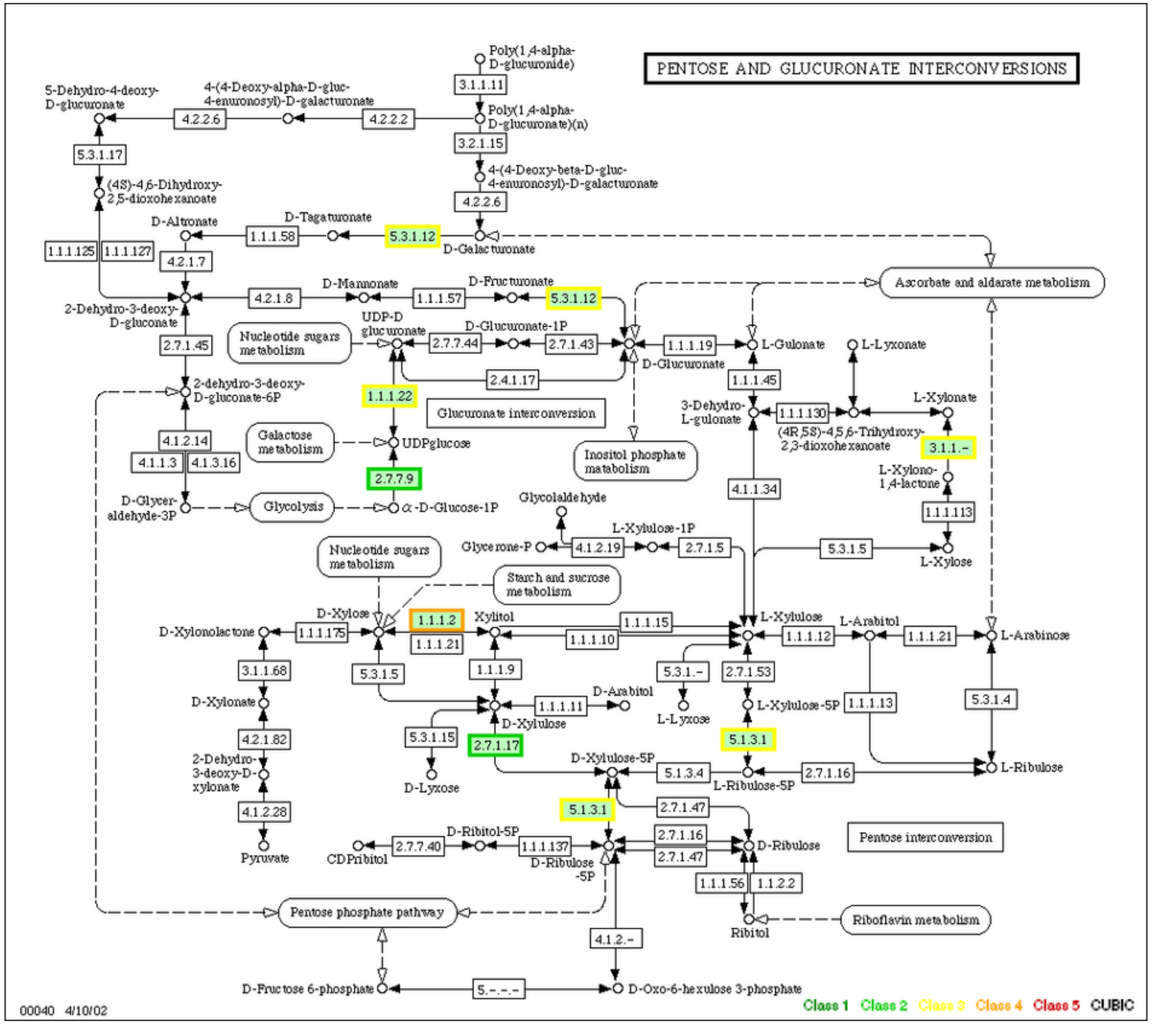


- Proteins
  - Structure prediction
  - Structure comparison
  - Protein dynamics





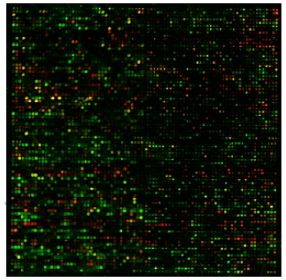
2.2 Biochemical Networks



2.2.1 Motivation

Network Biology

- Classical biology $\Rightarrow$ Learning from the individual
- Systems biology $\Rightarrow$ Biological processes are integrated systems of many diverse interacting components
- Starting with the human genome project, high-throughput methods generate a huge amount of data



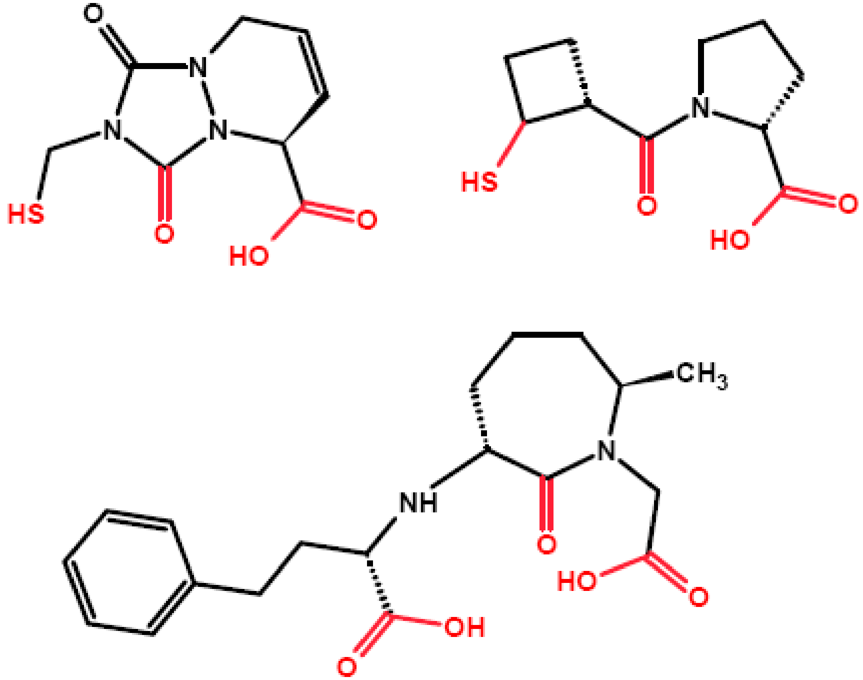
- Networks are <u>the</u> key concept to structure and combine that data

[Inspired by O. Kohlbacher's talk at Dagstuhl 08]

2.2.1 Motivation

There is a hierarchy of such biological networks

- (Molecular Graphs)
- Metabolic Networks
- Interaction Networks
- Regulatory Networks
- …



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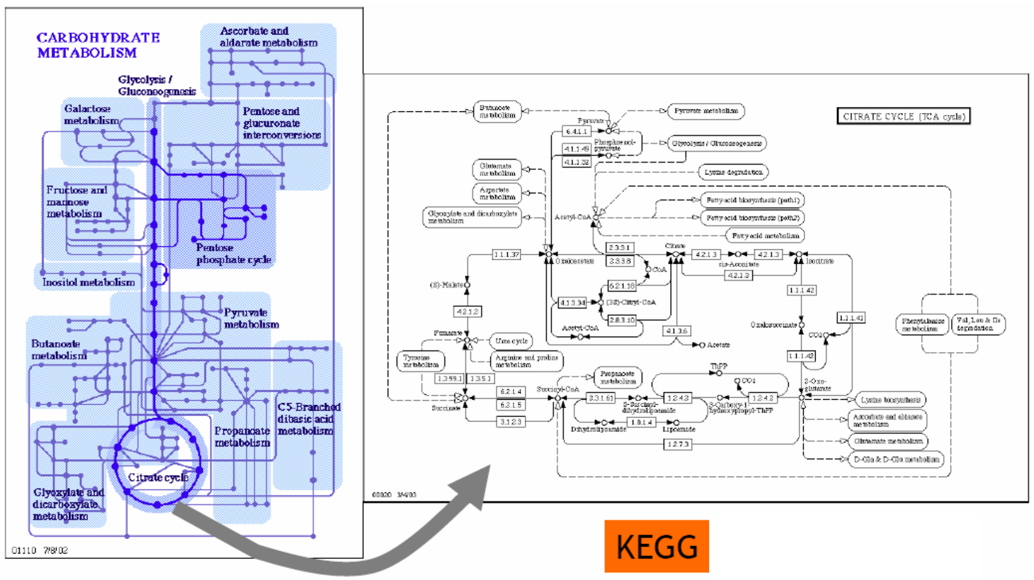

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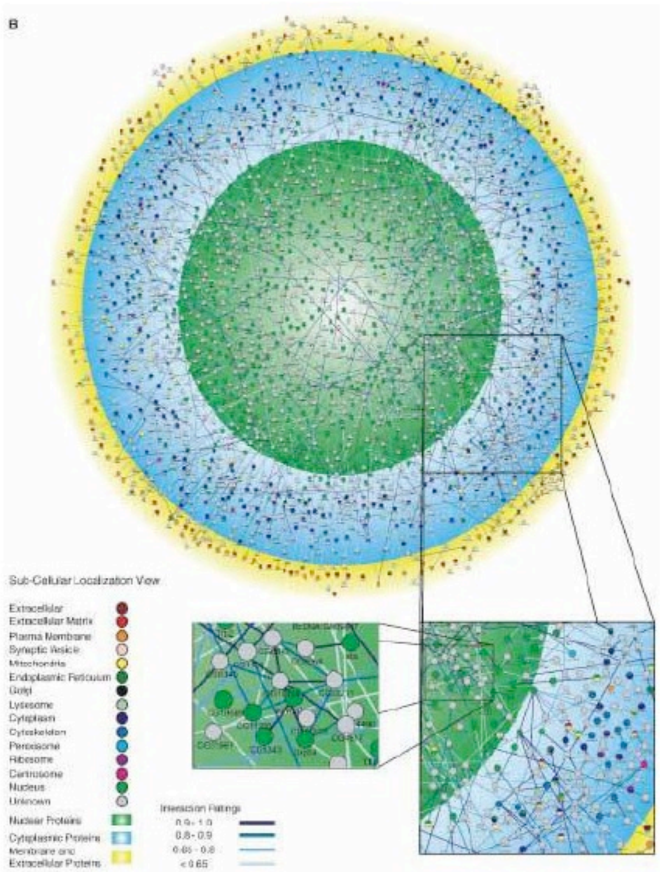


2.2.1 Motivation

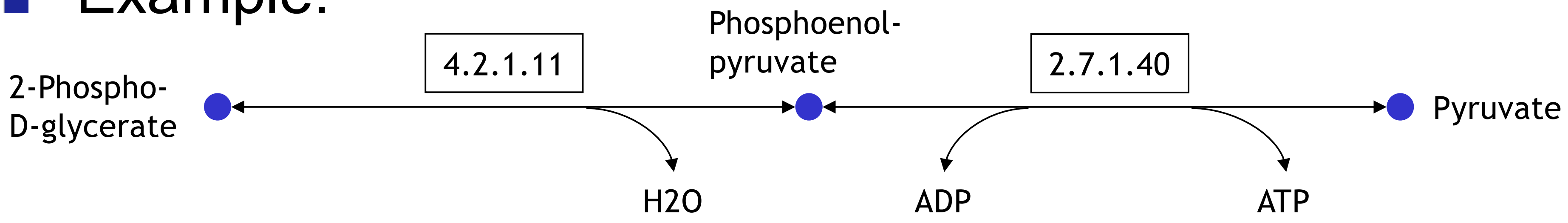
Why is that important?

- Diseases can be explained in context of networks
  - Infection: a foreign network starts operating in our own network
  - Genetic defects: incorrect connectivity of an element

- Applications
  - Drug design, metabolic engineering, …

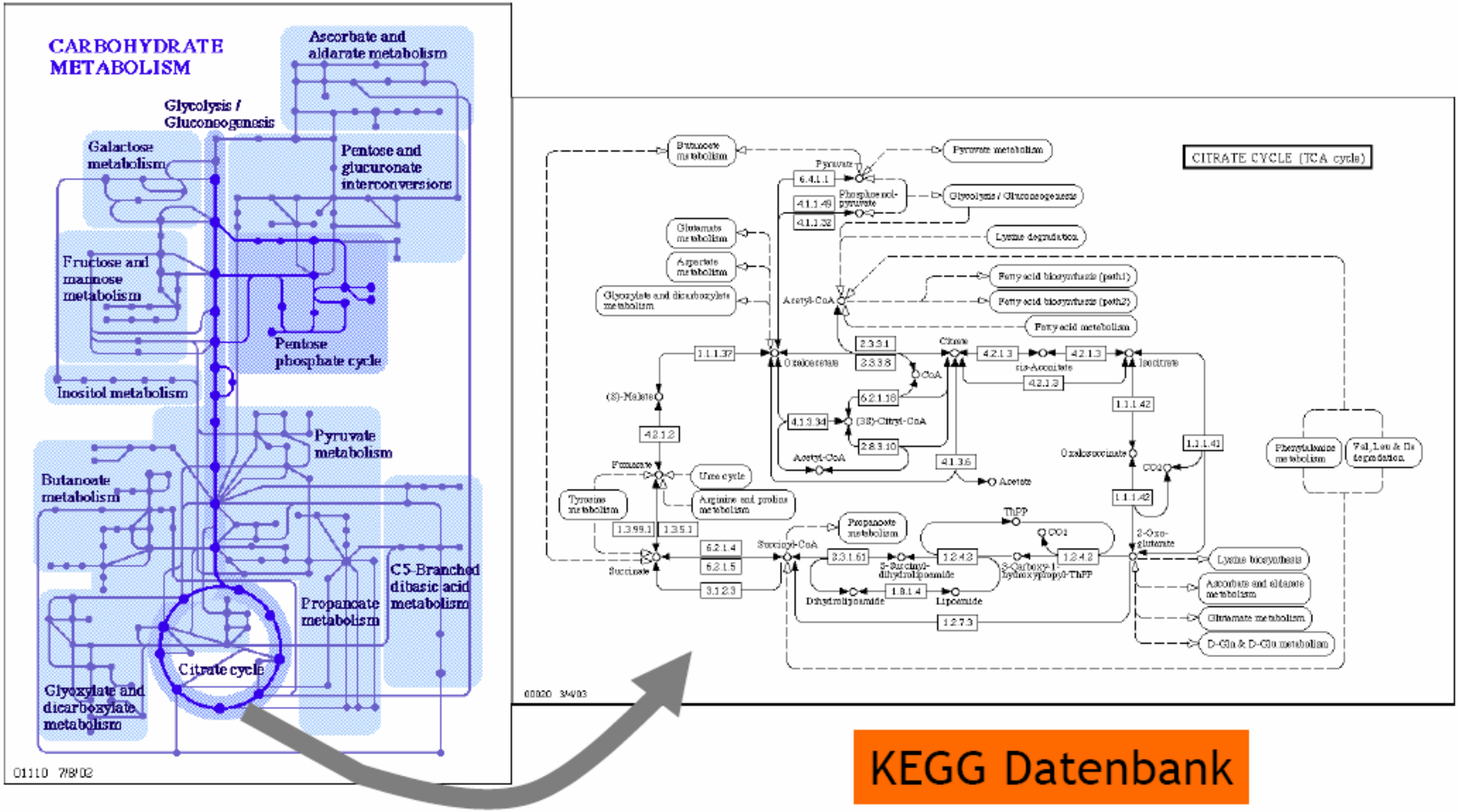
2.2.2 Metabolic Networks

- Paths: Sequence of reactions that provide all together for the translation of one substance into another one (pathways)
- One single reaction is defined by a catalyzing enzyme
- Example:



- As a whole, the paths build a so-called metabolic network (MN) [→ Graph]

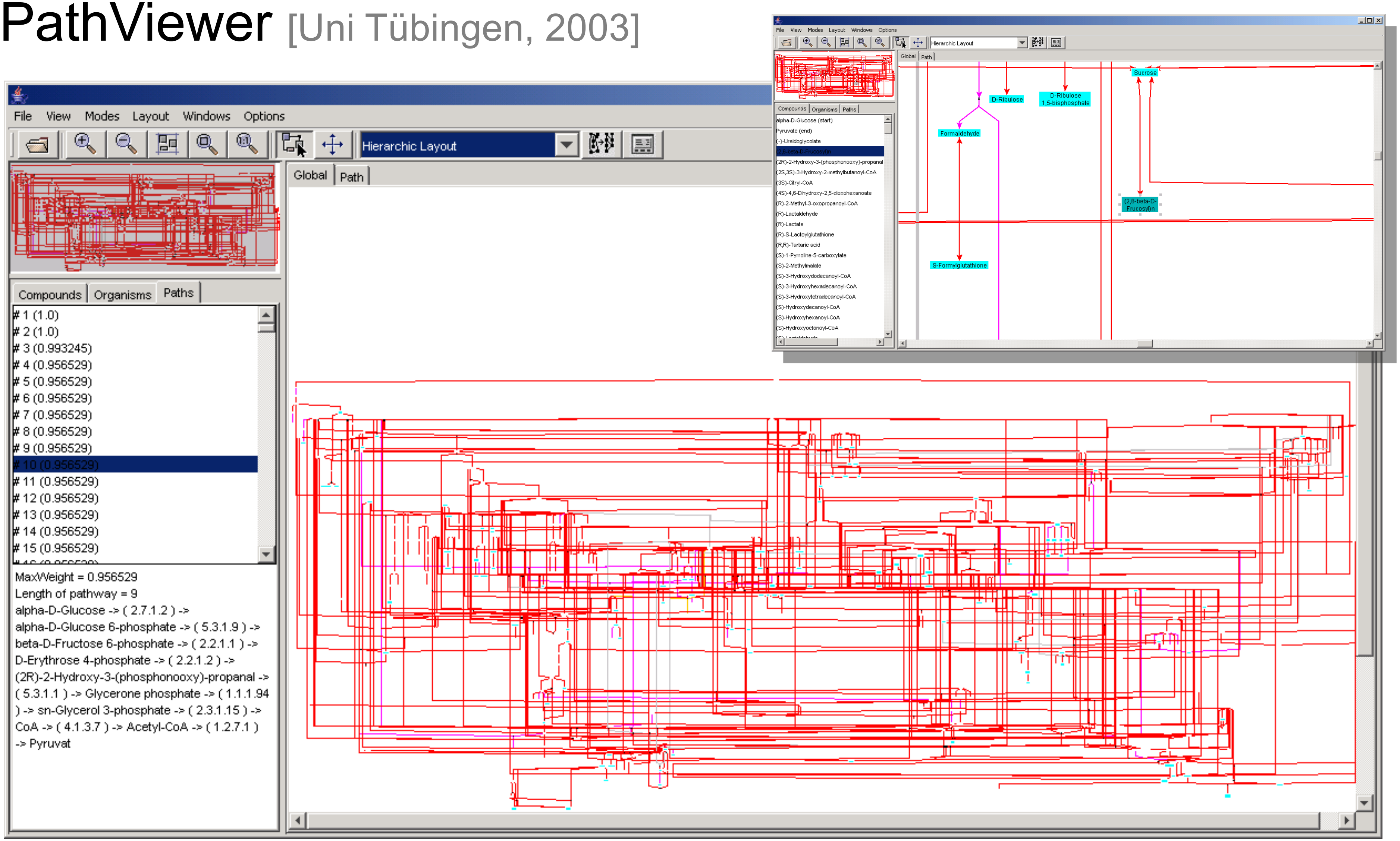
2.2.2 Metabolic Networks



KEGG Datenbank

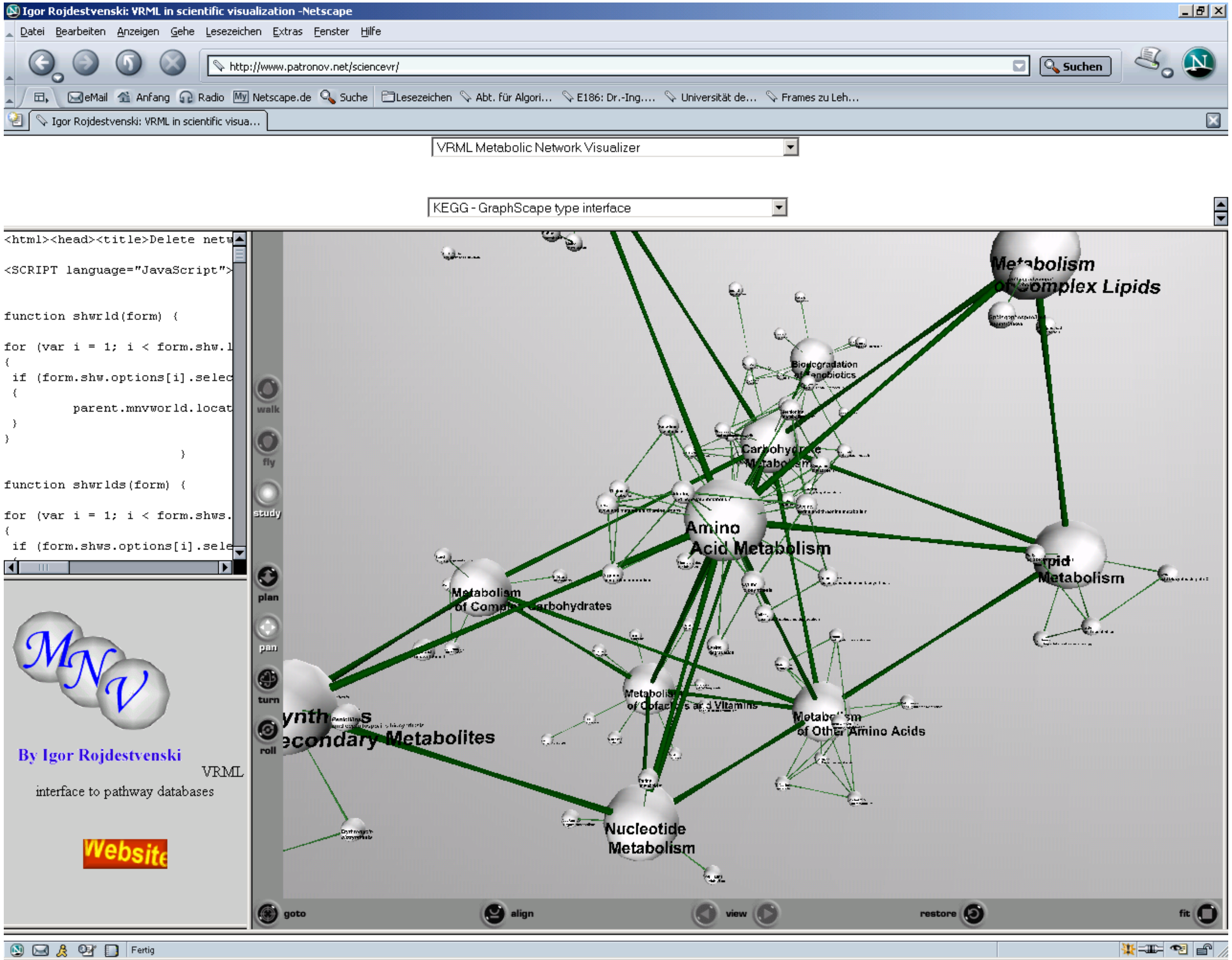
2.2.2 Metabolic Networks

PathViewer [Uni Tübingen, 2003]



2.2.2 Metabolic Networks

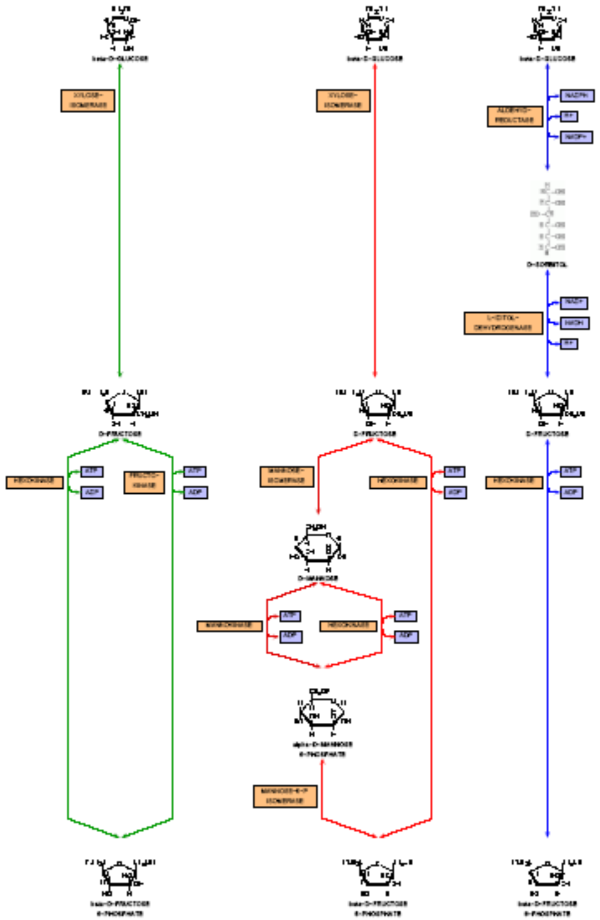
MNV [Rojdestvenski, 2003]

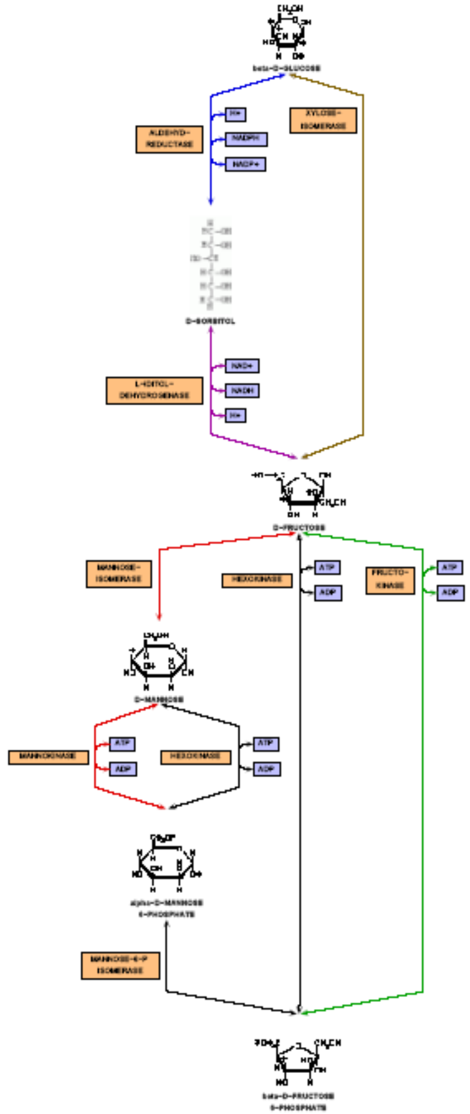


http://bioinformatics.oxfordjournals.org/content/19/18/2436.full.pdf

2.2.2 Metabolic Networks

BioPath [Schreiber, 2002]



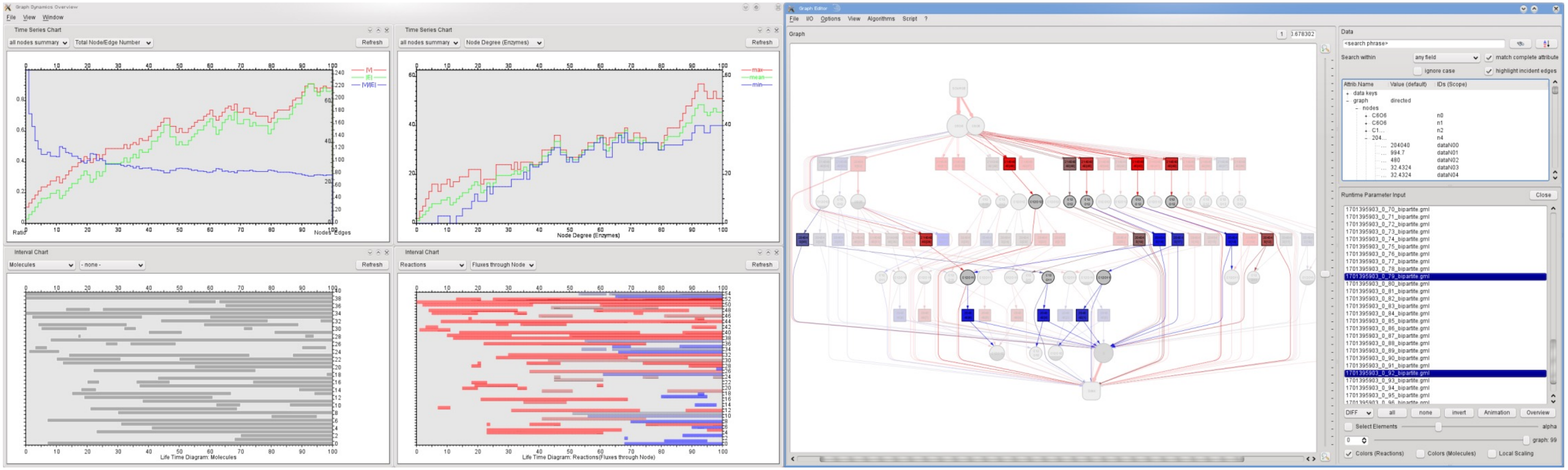


http://www.bioinfo.de/isb/2002/02/0006/

Visualization of Simulation Data

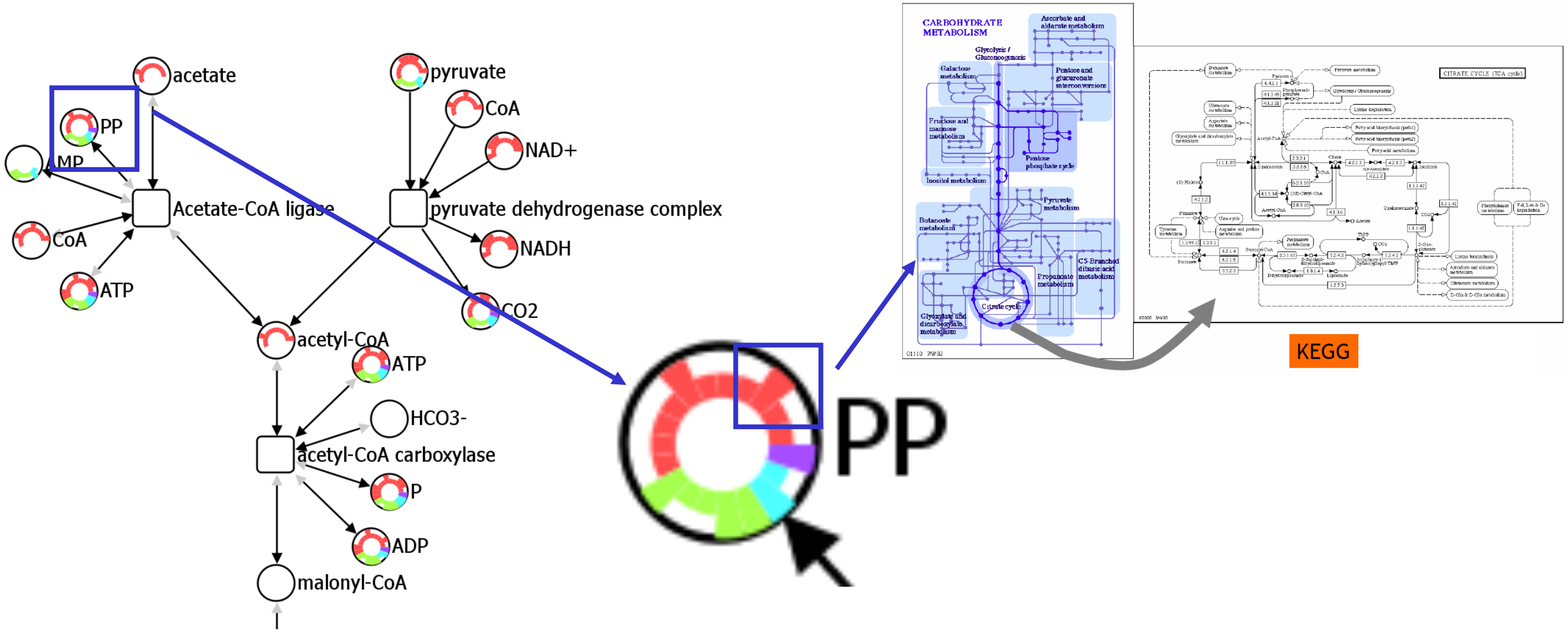
[Uni Leipzig & LNU, 2009]

[Video]



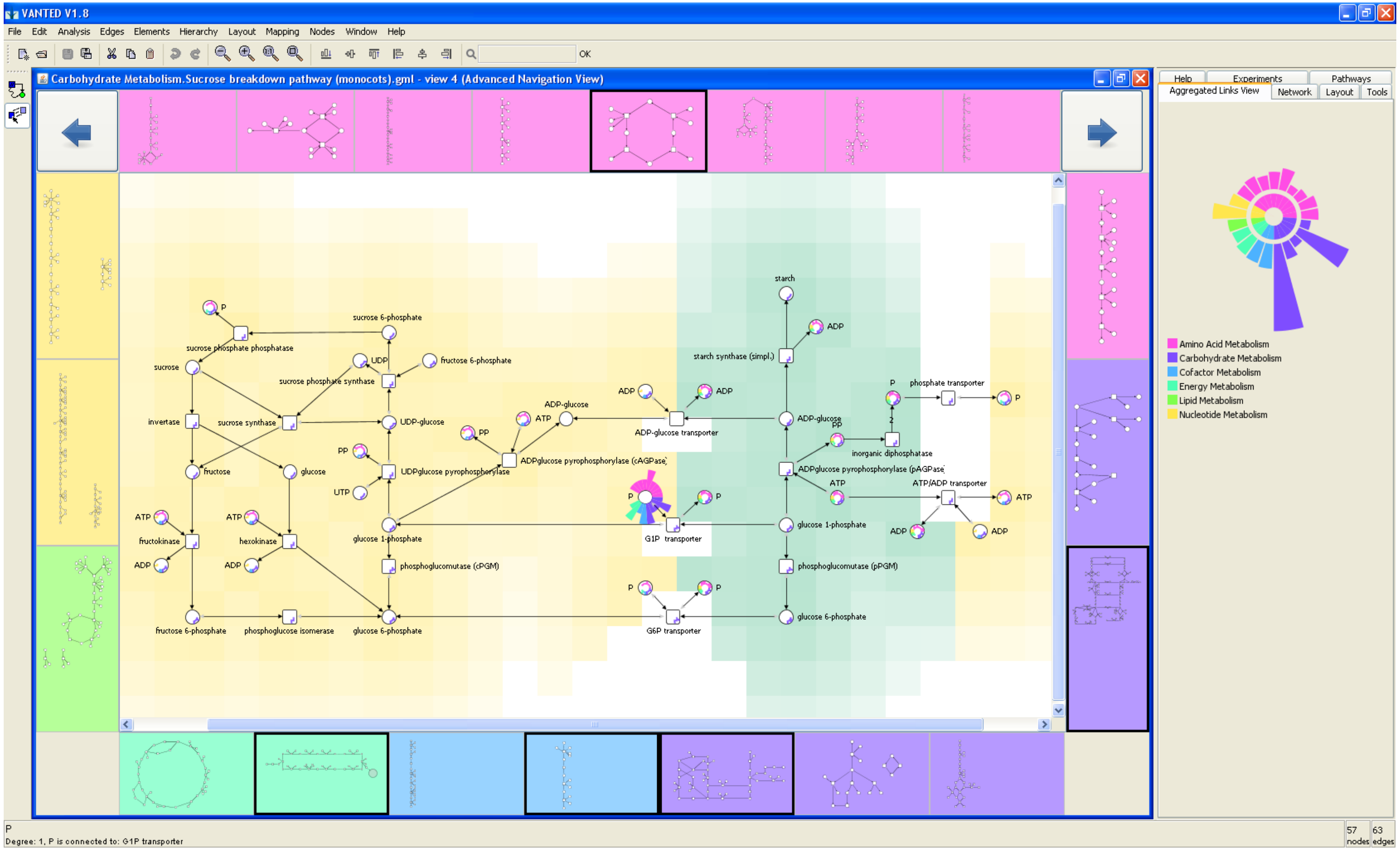
Interrelationships between network components

[LNU & IPG Gatersleben, 2012]



2.2.2 Metabolic Networks

[Video]



http://sourceforge.net/projects/gliep/

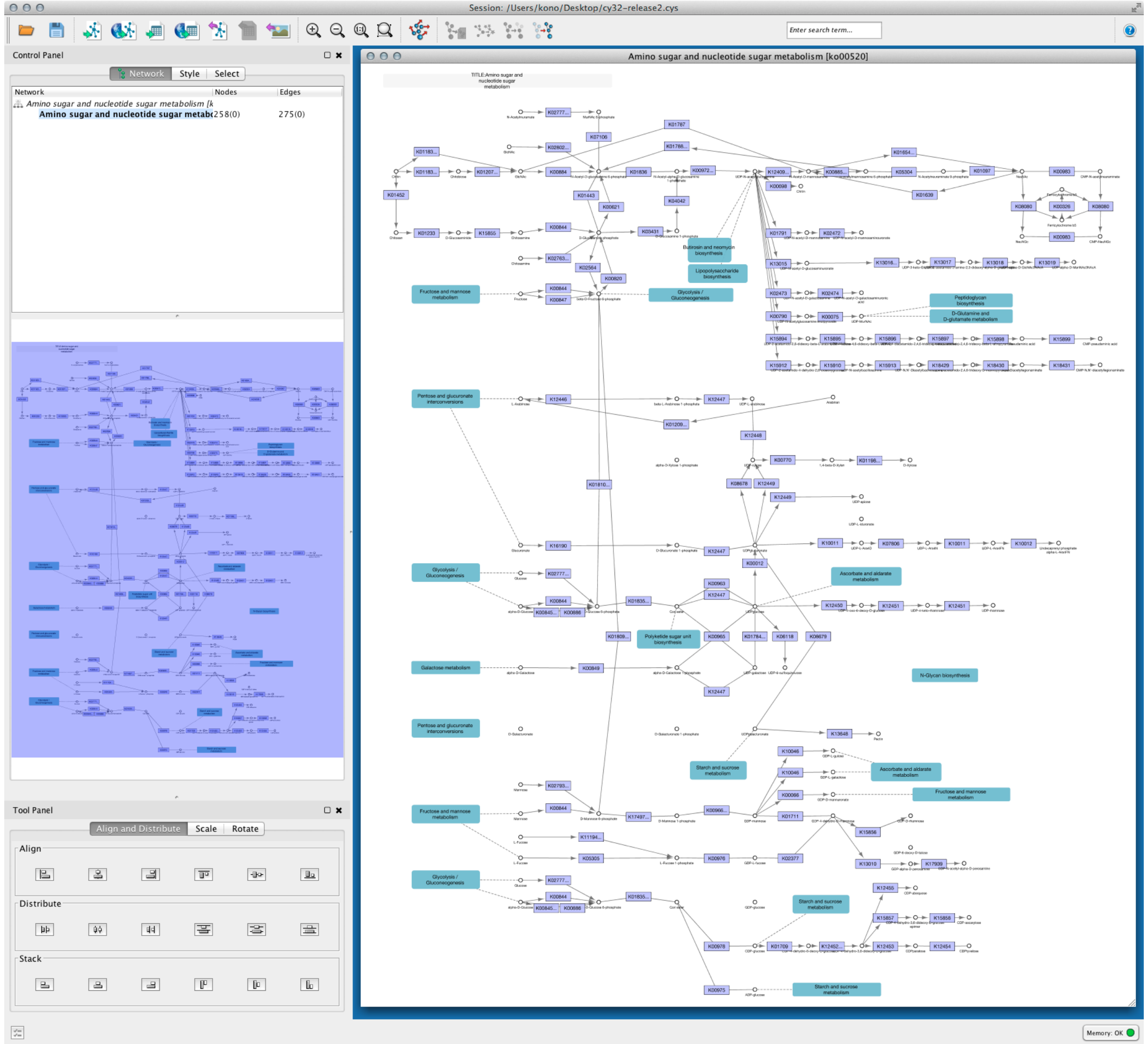
http://cs.lnu.se/isovis/pubs/kerren-ivi12-postprint.pdf

2.2.3 Protein-Protein Interactions

- Nodes are proteins and edges the interactions between them
- Interactions between proteins are important for many biological functions
- Example
  - Signal Transduction: signals from the exterior of a cell are mediated to the inside of that cell by protein-protein interactions of the signaling molecules
  - It plays a fundamental role in many biological processes and in many diseases

http://en.wikipedia.org/wiki/Protein-protein_interactions

Cytoscape

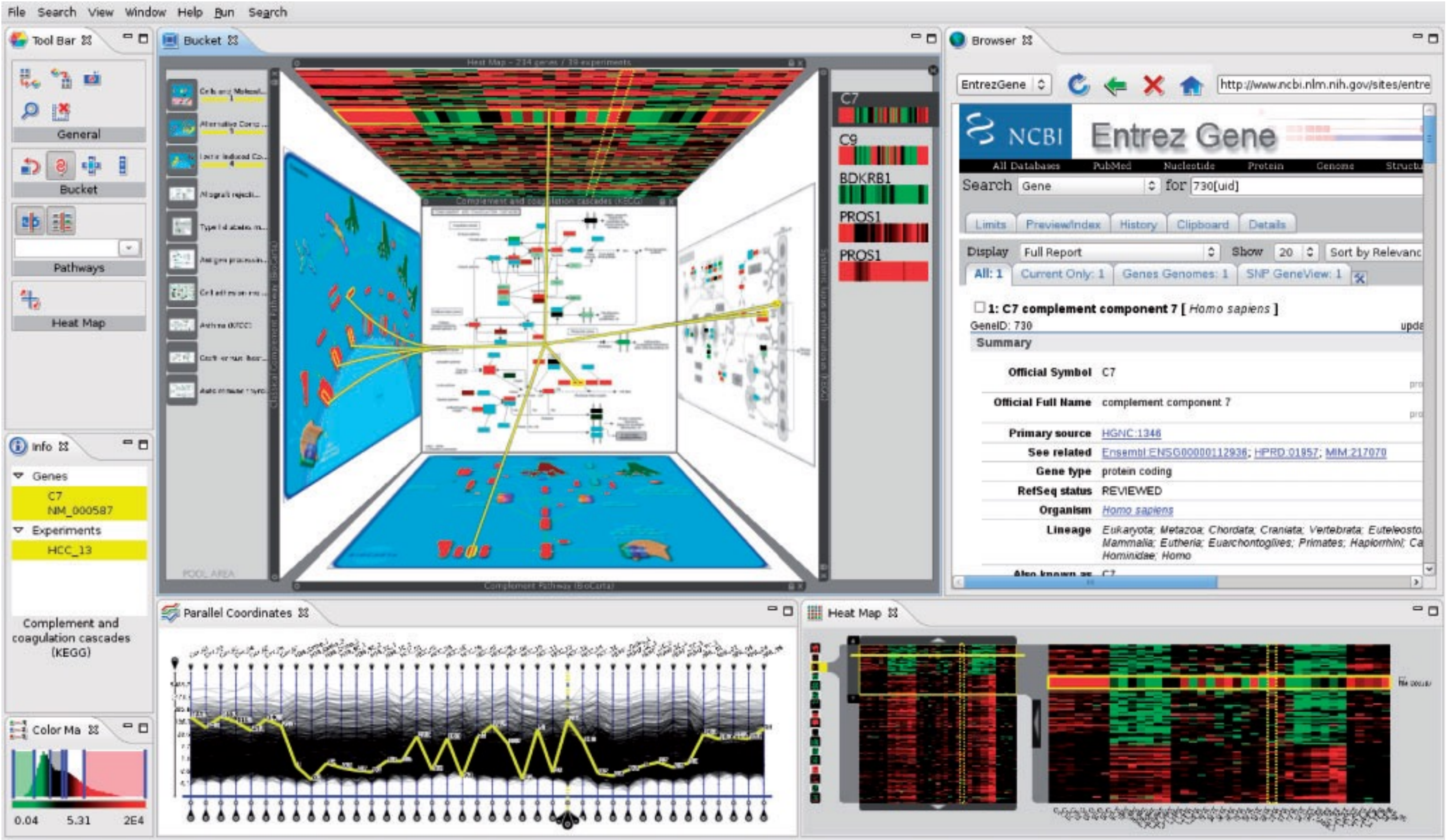


http://www.cytoscape.org

2.2.4 Special Visualizations

Showing relationships between pathways and other data

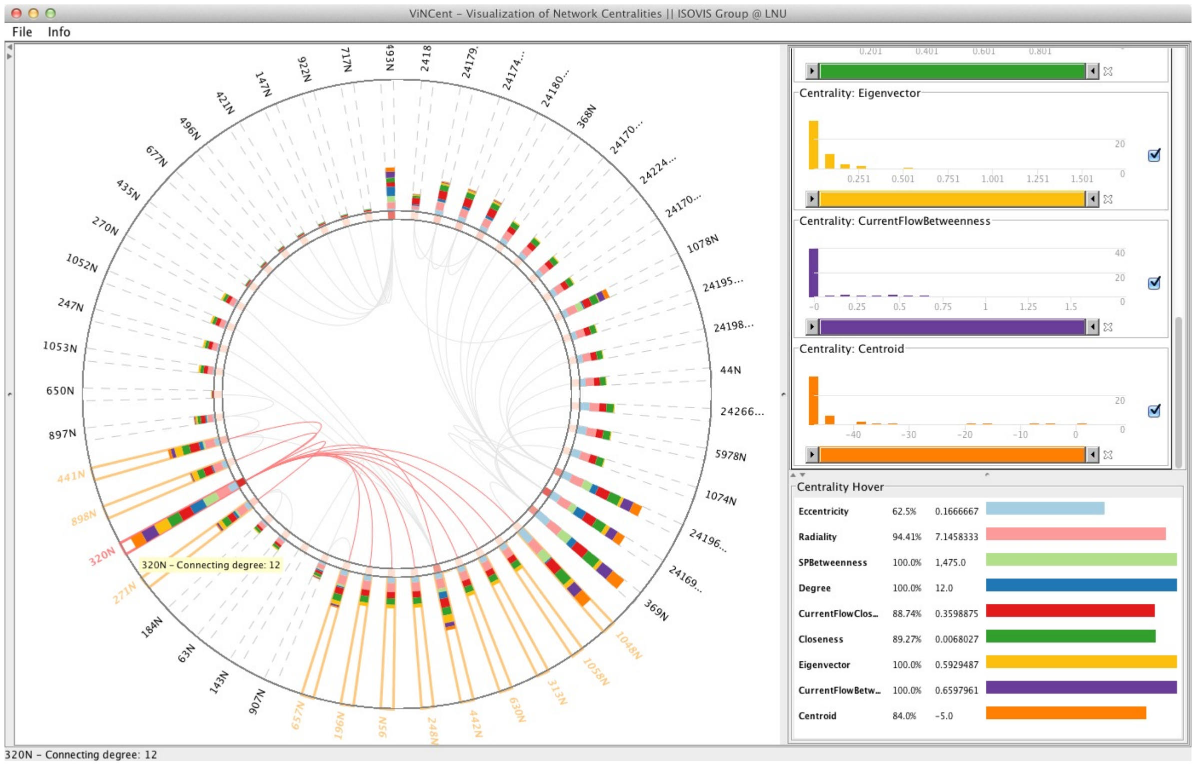
[Streit et al., 2009]



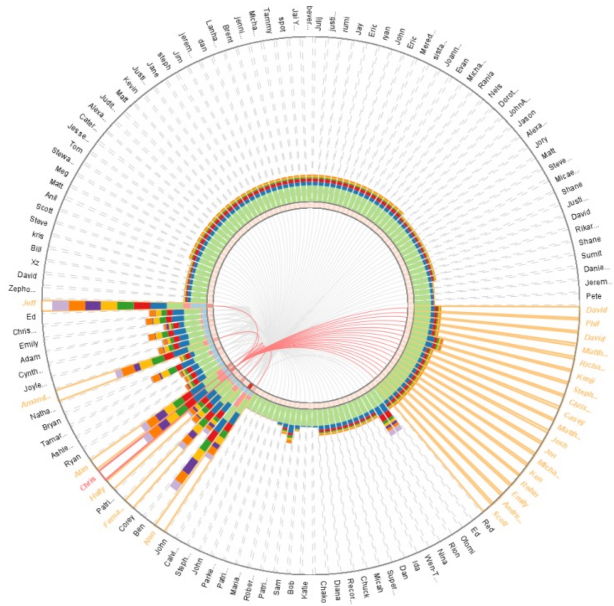
[Video]

2.2.4 Special Visualizations

- Visualization of Network Centralities [LNU, 2012]



[Video]



http://cs.lnu.se/isovis/pubs/docs/kerren-ivapp12-postprint.pdf

LNU

- Become an **exchange student** at Linnaeus University (LNU), Sweden
  - Many interesting courses in visualization, visual network/text analysis, etc.
  - Many open, research related theses projects at Master's and Bachelor's level
  - Nice city and environment

- More information
  - lnu.se